ABSTRACT OF THE DISCLOSURE

A system and method for ultra-high-bandwidth data transmission over short distances. The system comprises a transmitter unit containing a high-density array of individually addressable photon emitters and a receiver unit containing a corresponding array of photon sensors. The transmitter unit encodes digital data into a rapid sequence of frames, where each emitter's wavelength and intensity represents a data packet. The receiver unit captures these frames and decodes them back into digital data. The massive parallelism of millions of emitter/sensor pairs allows for data transfer rates several orders of magnitude greater than conventional electrical or fiber-optic cables, with the total bandwidth being a direct function of the number of elements in the arrays. While a preferred embodiment uses a micro-display and an image sensor operating in the visible light spectrum, the principle is extendable to non-visible spectra such as infrared or ultraviolet. An integrated-circuit-based negotiation protocol allows the receiver unit to calibrate itself to the specific characteristics of the transmitter array, optimizing precision and data fidelity.